Basic Principles of Medicine 1
Module: Cardiovascular, Pulmonary, Renal

Reactive Oxygen Species (ROS)

After: CPR lecture: pentose phosphate pathway

DLA Video

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DLA: Reactive Oxygen Species

SOM.MK.I.BPM1.1.CPR.2.BCHM. 0270	Define the terms radical, free radical and reactive oxygen species (ROS). Discuss cellular damage and effects of ROS.
SOM.MK.I.BPM1.1.CPR.2.BCHM. 0271	Describe formation of superoxide anion radical related to the ETC or cytochromes.
SOM.MK.I.BPM1.1.CPR.2.BCHM. 0272	Describe the reactions catalyzed by superoxide dismutase, catalase and glutathione peroxidase (selenium containing) and indicate the ROS scavenged.
SOM.MK.I.BPM1.1.CPR.2.BCHM. 0273	Discuss the formation of hydroxyl radicals related to ionizing radiation, Haber- Weiss reaction and Fenton reaction.
SOM.MK.I.BPM1.1.CPR.2.BCHM. 0274	Discuss the compartmentation of free radical defense in the cell.
SOM.MK.I.BPM1.1.CPR.2.BCHM. 0275	Indicate non-enzymatic radical scavengers. Include vitamin C and E (found in fruits and vegetables).
SOM.MK.I.BPM1.1.CPR.2.BCHM. 0276	Indicate the biochemical defect in amyotrophic lateral sclerosis (ALS).
SOM.MK.I.BPM1.1.CPR.2.BCHM. 0277	Discuss the beneficial role of ROS formation as an immunological response in phagocytosis and describe the reactions catalyzed by NADPH oxidase and myeloperoxidase.
SOM.MK.I.BPM1.1.CPR.2.BCHM. 0278	Discuss the direct and potential effects of excess nitric oxide formation. Discuss the importance of inducible nitric oxide synthase for formation of RNOS.
SOM.MK.I.BPM1.1.CPR.2.BCHM. 0279	Predict the effect of NADPH oxidase deficiency and myeloperoxidase deficiency. Indicate the biochemical defect in chronic granulomatous disease.

Recommended Reading for Reactive Oxygen Species

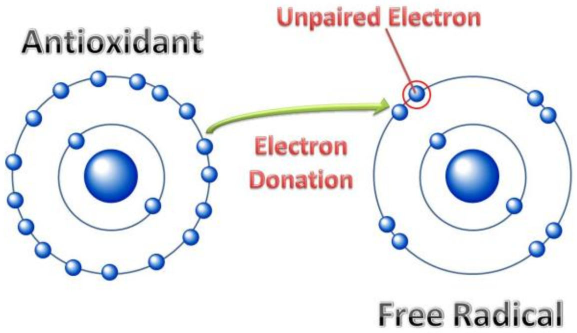
- **Lippincott's Illustrated Reviews Biochemistry 8th Ed; Pages 163-166** Pentose Phosphate Pathway and Nicotinamide Adenine Dinucleotide Phosphate | Lippincott® Illustrated Reviews: Biochemistry, 8e | Medical Education | Health Library (lwwhealthlibrary.com)
- Mark's Basic Medical Biochemistry 6th edition Oxygen Toxicity and Free-Radical Injury | Marks' Basic Medical Biochemistry: A Clinical Approach, 6e | Medical Education | Health Library (lwwhealthlibrary.com) MBMB
- http://www.ncbi.nlm.nih.gov/pmc/articles/PMC3166784/ ; Hydroxyl radical and its scavengers in health and disease. B.Lipinski, Oxid Med Cell Longev. 2011: 809696

Reactive Oxygen Species (ROS)

Dr. Margit Trotz, Ph.D.
Emeritus Professor

What are radicals and what are ROS?

A **free radical** is a molecule that has a single unpaired electron in an orbital and is highly reactive.



Free radicals can initiate chain reactions by extracting an electron from another molecule and they are scavenged by antioxidants.

ROS is a group of molecules that is constantly formed during aerobic metabolism and includes:

$O_2 + e^- \rightarrow$ superoxide $O_2^{\cdot -}$

$\downarrow + e^-$

hydrogen peroxide H_2O_2

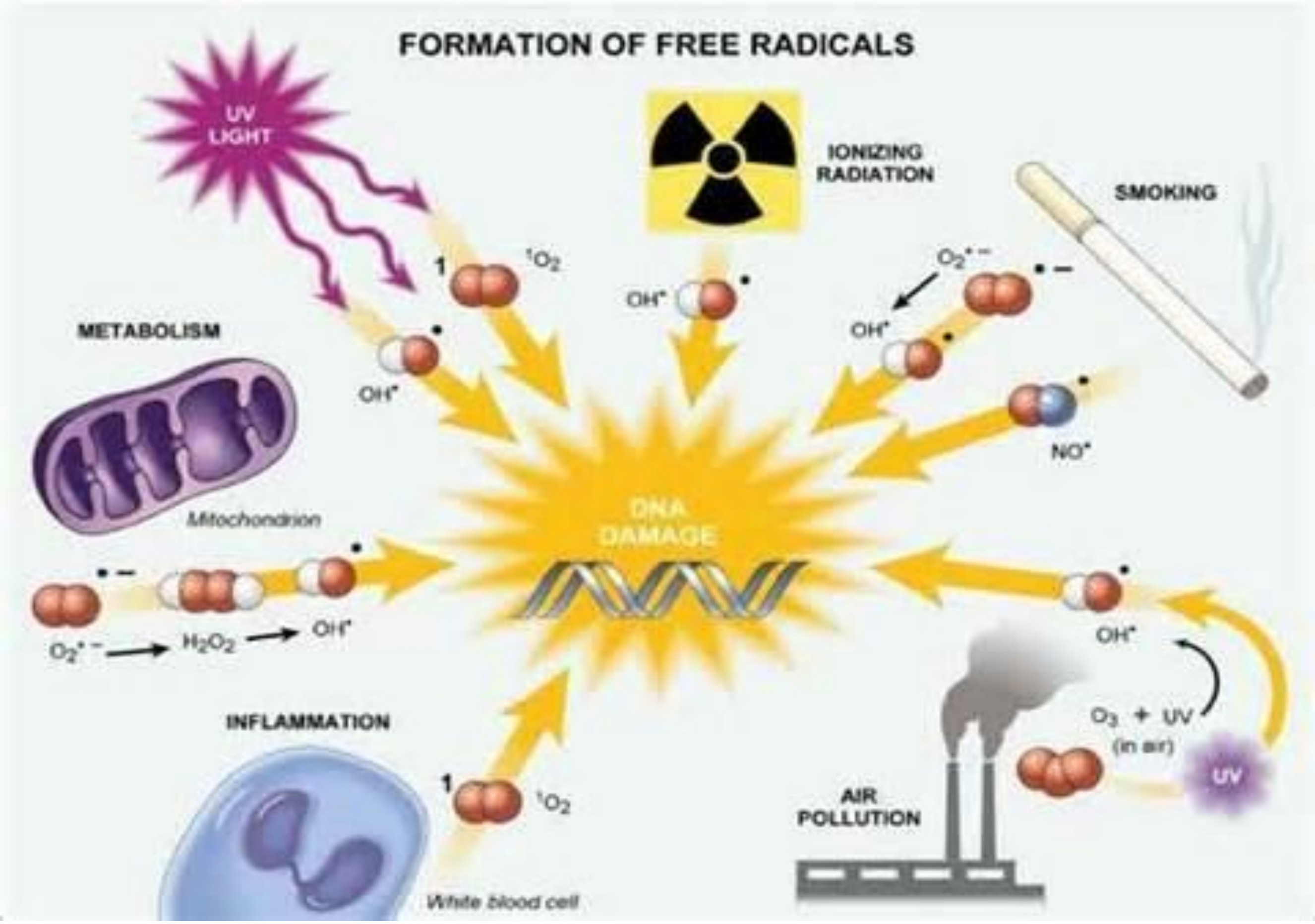
$\downarrow + e^-$

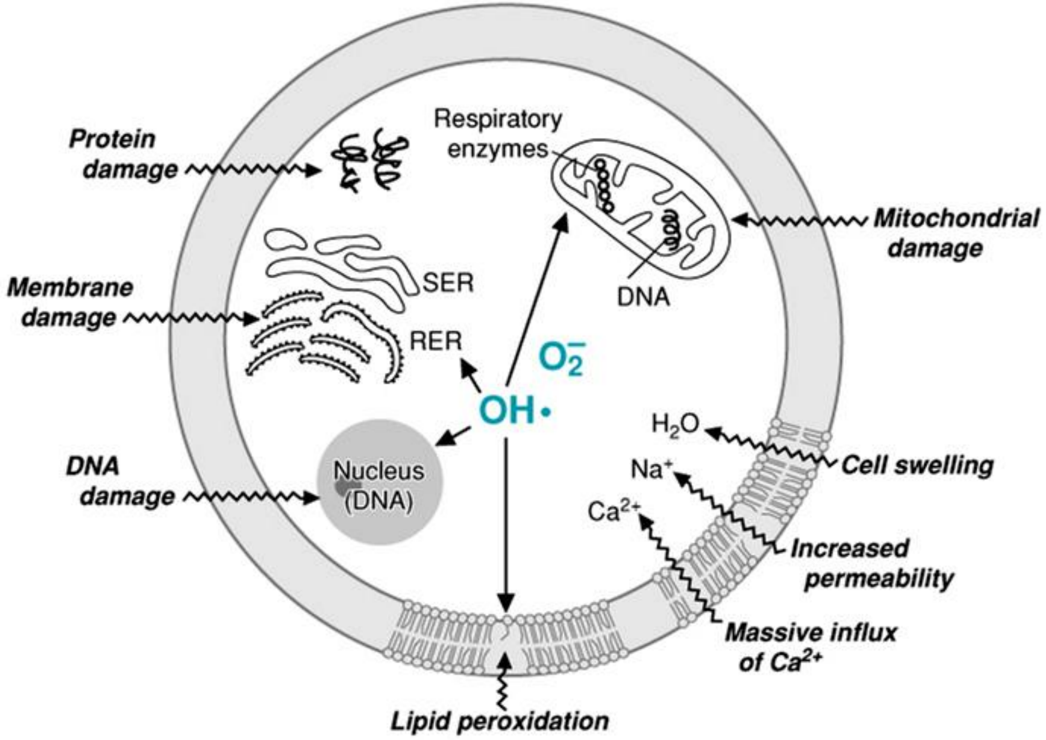
hydroxyl radical $OH^{\cdot}$ $+e^- \rightarrow H_2O$

ROS are highly reactive and cells have several protective systems to scavenge these molecules and prevent toxicity.

ROS can damage DNA, lipids and proteins.

This can lead to cancer, reperfusion injury, inflammation, atherosclerosis and aging.





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Superoxide results from molecular oxygen after uptake of one electron (superoxide anion radical)

The **electron transport chain (ETC)** is in the inner mitochondrial membrane. The transported electrons are prevented to react with oxygen until they reach the last step of the ETC where water is formed.

At the level of **Coenzyme Q**, one electron can escape, bind to molecular oxygen and form superoxide.

Cytochromes P450 are enzymes that use molecular oxygen and form often a hydroxyl group, they are monooxygenases. Radicals are generated in this complex process and superoxide is formed.

Cytochromes P450 are used for drug metabolism and in different organs for the synthesis of bile acids, steroid hormones and formation of 1,25-dihydroxy vitamin D.

Enzymatic scavenging of ROS

ROS are scavenged by 3 key enzymes:

1. **Superoxide dismutase** acts superoxide
2. **Catalase** acts on hydrogen peroxide
3. **Glutathione peroxidase** acts on hydrogen peroxide and organic peroxides

Note: Hydroxyl radical cannot be scavenged by enzymes, it is too reactive. It is scavenged by antioxidants.

1. Superoxide dismutase (SOD)

uses **2 superoxides** as **substrates**
and forms
hydrogen peroxide and **oxygen** as **products.**

$$O_2^{\cdot -} \;+\; O_2^{\cdot -}$$

$$2\,H^+$$

$$H_2O_2 + O_2$$

2. Catalase

uses **2 hydrogen peroxides** as **substrates**
and forms
water and **oxygen** as **products.**

$$H_2O_2 \;+\; H_2O_2$$

$$2\,H_2O \;+\; O_2$$

3. Glutathione peroxidase

uses **one hydrogen peroxide and an** organic peroxide as **substrate** and forms **water** and **oxygen** as **products. Uses** two molecules of reduced glutathione (GSH) and forms GSSG.

Contains selenium (antioxidant)

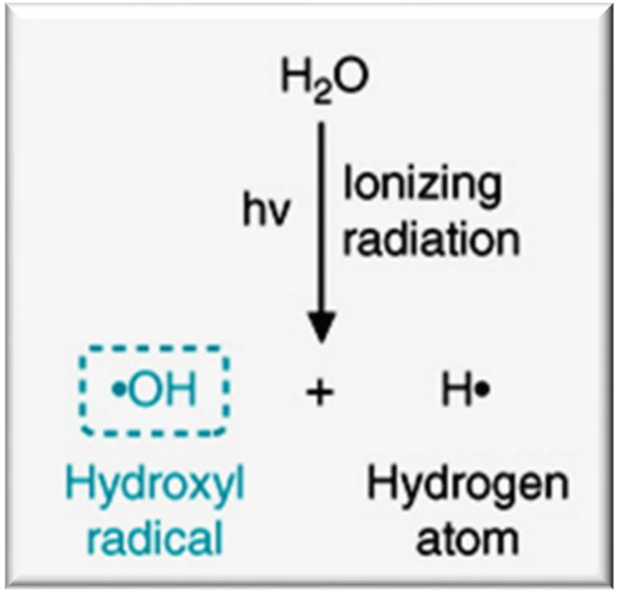
H_2O_2 plus 2 GSH → 2 H_2O plus GSSG

ROOH plus 2 GSH → ROH plus GSSG

Hydroxyl radical OH·
is the most aggressive ROS.

Hydroxyl radical is formed from water during ionizing radiation.

Radiation: damage to the skin, mutations, cancer and death.



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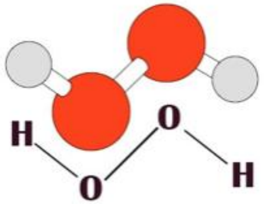
Hydrogen peroxide can lead to hydroxyl radicals



H_2O_2 can **diffuse** through **membranes.**

It can nonenzymatically react with superoxide or ferrous iron

which leads to formation of **hydroxyl radicals.**

Hydroxyl radicals are formed in the :

1. Haber-Weiss reaction

2. Fenton reaction

Hydroxyl radical $OH^{\cdot}$
is formed from hydrogen peroxide

HYDROGEN PEROXIDE and superoxide: (Haber-Weiss reaction)

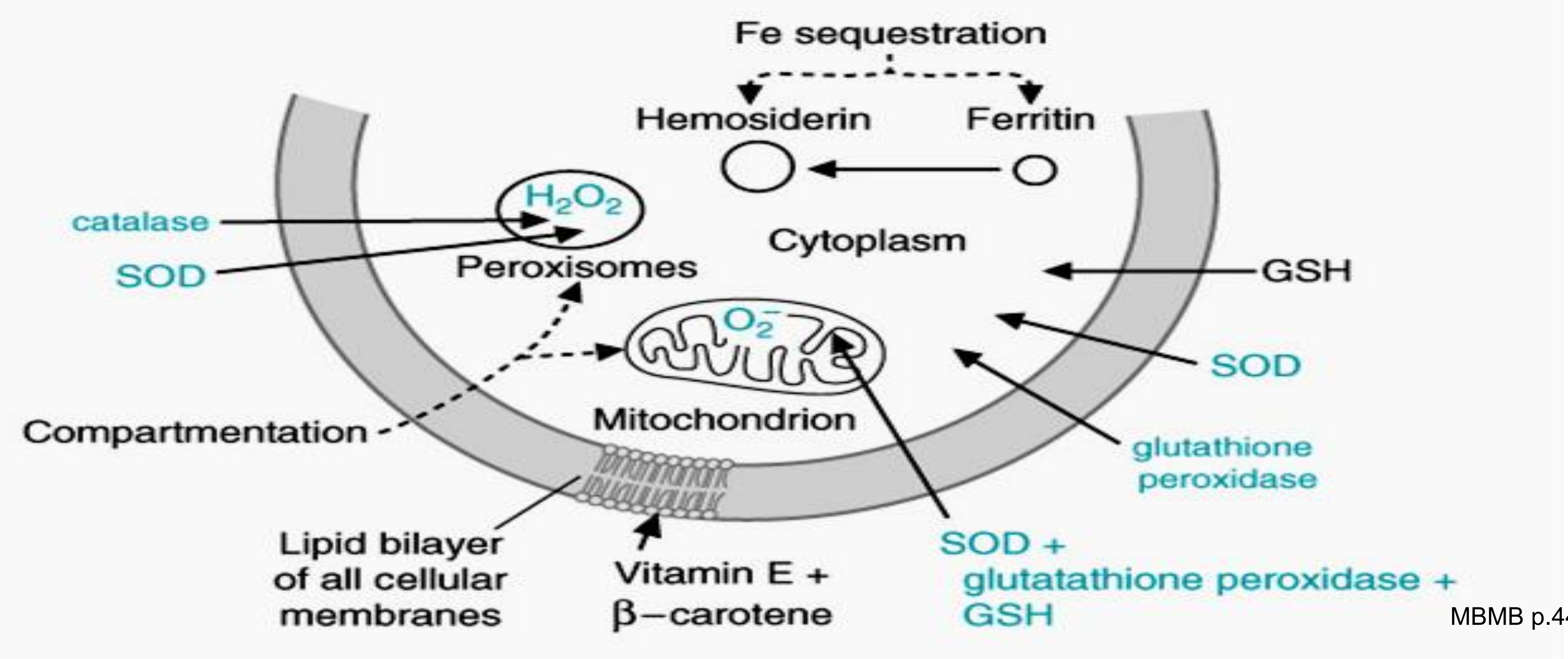
$$H_2O_2 + O_2^{\cdot -} + H^+ \downarrow OH^{\cdot} + H_2O + O_2$$

HYDROGEN PEROXIDE and ferrous iron: (Fenton reaction)

$$H_2O_2 + Fe^{2+} \downarrow OH^{\cdot} + OH^- + Fe^{3+}$$

Compartmentation of free radical defenses

SOD and glutathione peroxidase isozymes are found in cytosol and mitochondria. Iron is sequestered and stored safely in ferritin. Vitamins are protecting membranes and GSH is also an antioxidant.



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Radical scavengers

1. Enzymes:

Superoxide dismutases, catalase and glutathione peroxidase.

Cofactors needed are copper, zinc, manganese, iron and selenium.

2. Endogenous natural scavengers:

Uric acid, bilirubin, glutathione and other

3. Dietary radical scavengers

Vitamins C and E, carotenoids, flavonoids, phytochemicals, polyphenols, isoprenoids, PUFA and many more.

Deficiency of superoxide dismutase (SOD) leads to
Amyotrophic Lateral Sclerosis (ALS)

ALS is also known as **Lou Gehrig's** disease. He was a famous baseball player who became ill and passed away.

90% of the time the cause of superoxide dismutase deficiency is **idiopathic**. In the remaining 10% the cause is due to a **hereditary defect** of the gene for **superoxide dismutase.**

ALS is a **deadly degenerate disease** of upper and lower **motor neurons.** Onset mostly after 40 – 50 years of age.

Symptoms:

- **Difficulty breathing**, **vocal cord** dysfunction.
- **Head drop** due to **weakness** of the **neck muscles.**
- **Muscle weakness**, twitching, atrophy, dysphagia, hyperreflexia

Immune defense

ROS

RNOS

There are **situations** where the **body naturally** and on purpose **forms** ROS and RNOS as an **immune defense.**

This **occurs** in the **phagocytic cells:**
neutrophils, monocytes/macrophages and eosinophils

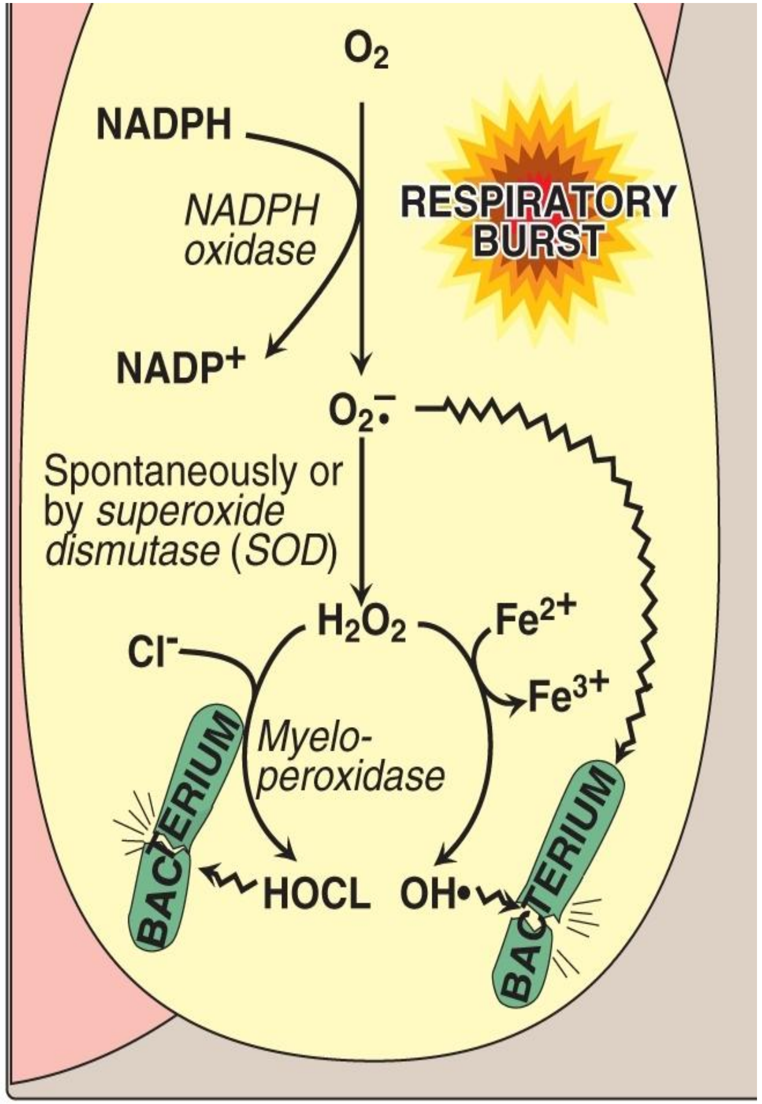
Key enzymes in phagolysosomes

1. **NADPH oxidase generates** on purpose **superoxide** using molecular oxygen and NADPH (respiratory burst) in phagolysosomes.

2. **Superoxide dismutase** forms $\mathbf{H_2O_2}$.

3. **Hydroxyl radicals** are formed in the Fenton reaction.

4. **Myeloperoxidase** is secreted into the phagolysosome and forms **hypochlorous acid** (bleach) which destroys **bacteria** and **fungi**.

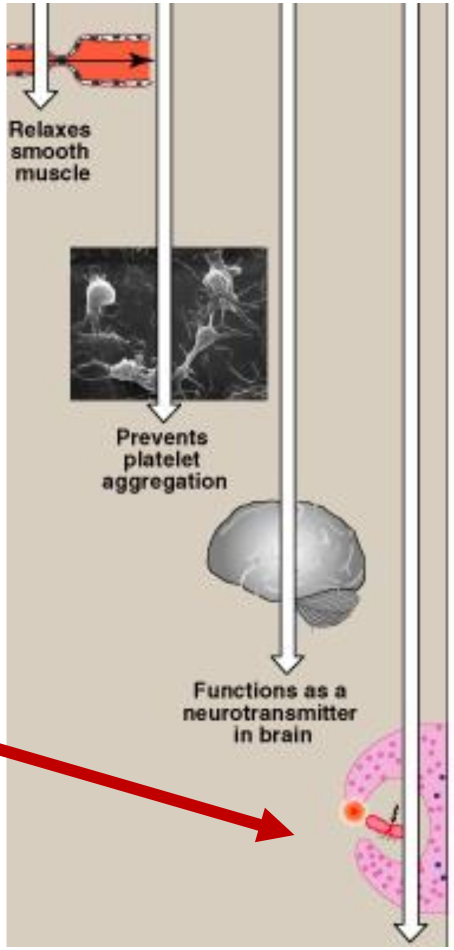


Nitric oxide (NO) leads to RNOS

Physiological functions of nitric oxide

- Relaxes smooth muscle (vasodilation).
- Prevents platelet aggregations.
- Functions as neurotransmitter in brain.
- **Destruction** of **invading bacteria.**

Nitric oxide formed in **macrophages** by **inducible NO synthase (iNOS)** is used for **RNOS formation** as **cell defense.**

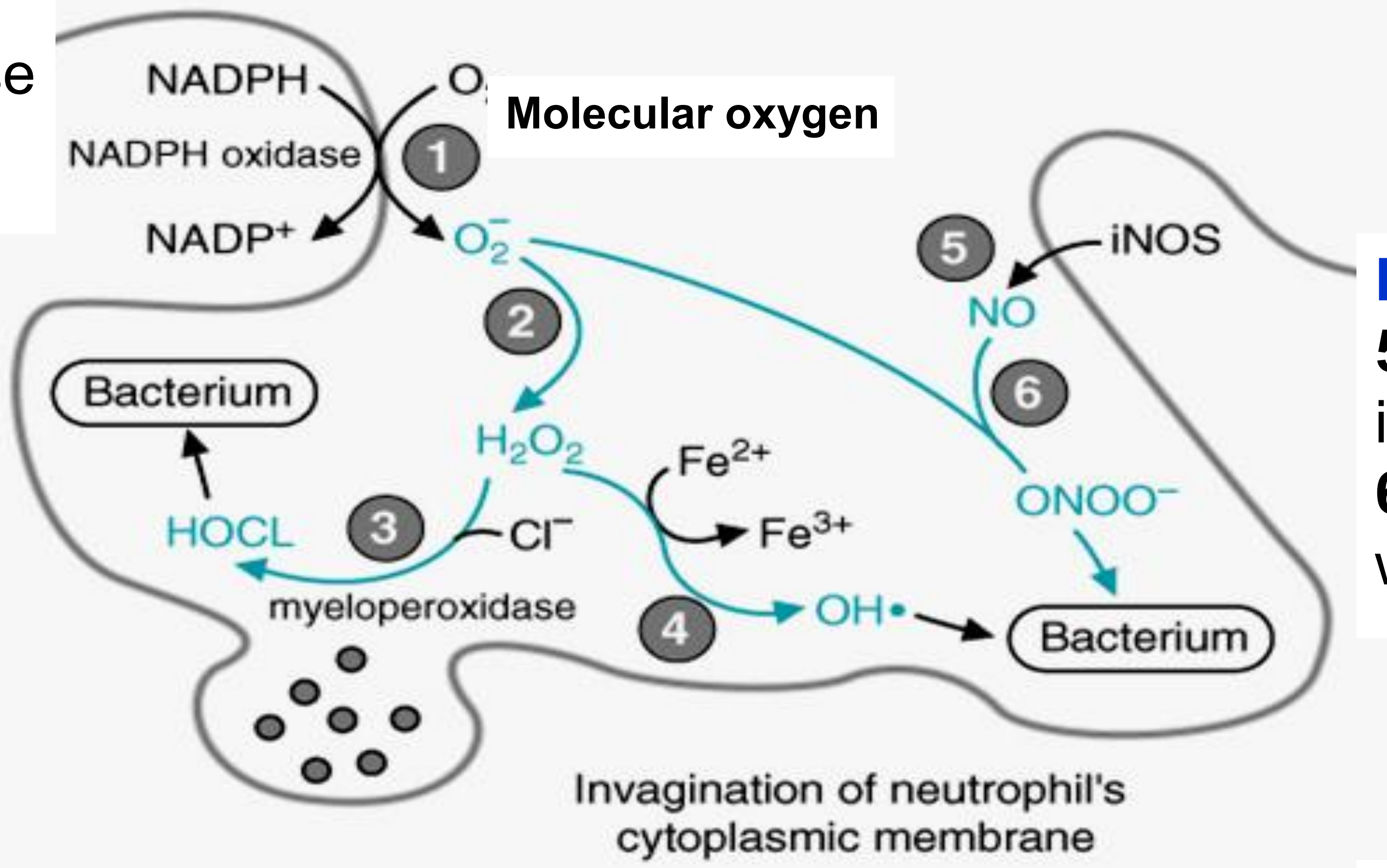


Note: **sepsis** can lead to **excessive NO** levels, **vasodilation** and severe **hypotension**.

Summary: **Formation of ROS, HOCL and RNOS**

ROS key reactions:

1.NADPH oxidase
2.SOD
3.Myeloperoxidase
4.Fenton reaction



RNOS:

5. NO formed by iNOS
6. reacts with superoxide

Intracellular

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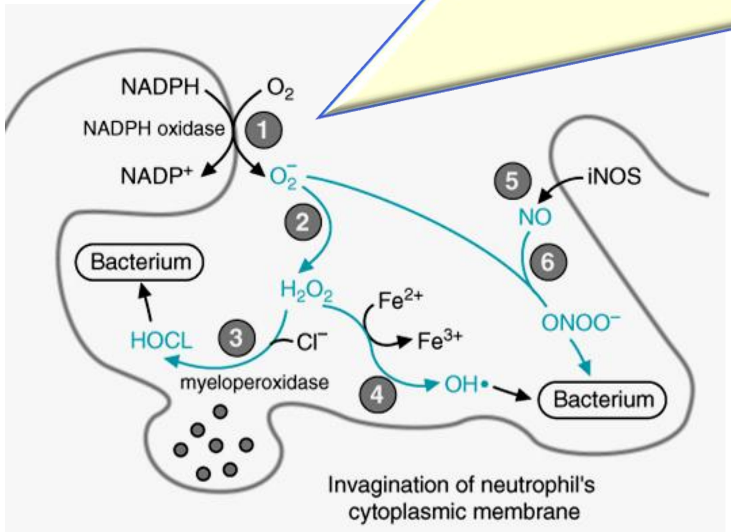
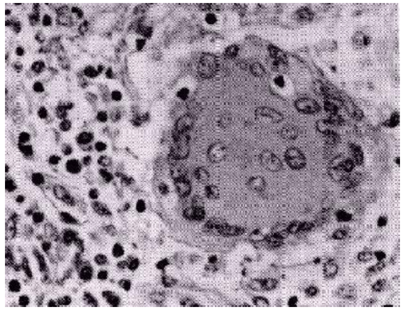
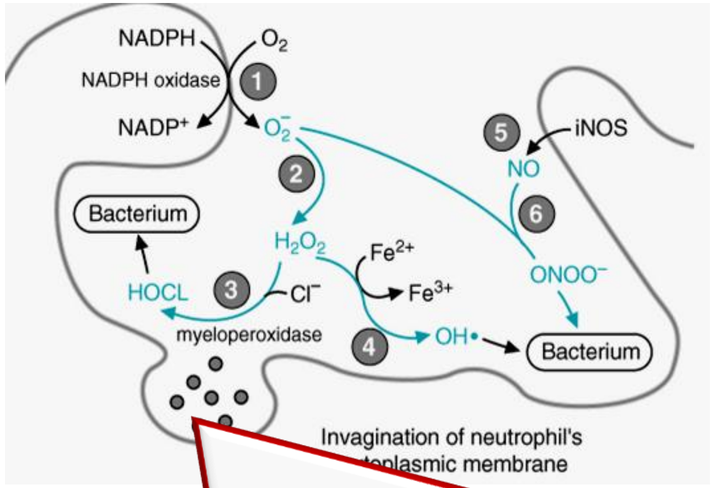
Chronic Granulomatous Disease (CGD)
Hereditary NADPH oxidase deficiency

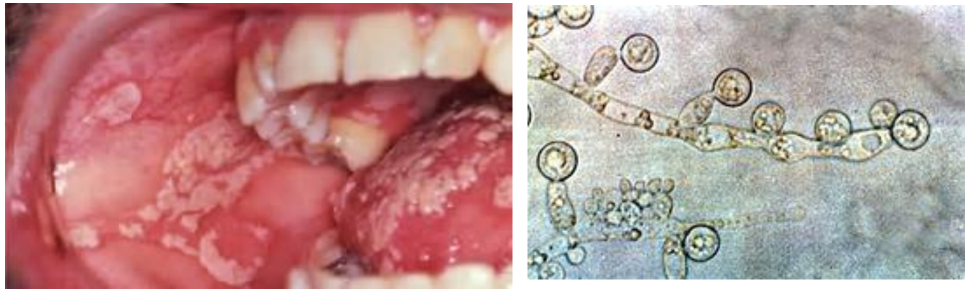




Granulomas contain sequestered **bacteria** in **infected cells.**

- **Recurrent** severe **infections** by **bacteria** (pneumonia) or **fungi** (often aspergillus)
- **Respiratory burst test** is **negative: less** ROS, RNOS and HOCL.
- Mostly **diagnosed** before **5 yrs old.**
- Most of the **genetic defects** are **X-linked.**





Candida albicans

Myeloperoxidase deficiency
Lack of hypochlorous acid

Recurrent severe infections by **candida albicans.**

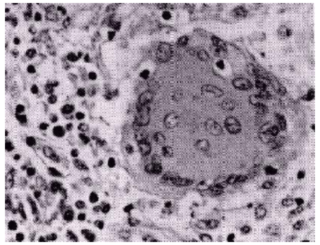
Decreased **formation** of **HOCL (bleach)** leads to

1. Oral and **genital infections.**

2. Systemic infections

Note: Bacteria are still destroyed as superoxide and nitric oxide are formed.

Chronic granulomatous disease (CGD)



NADPH oxidase Deficiency
(respiratory burst test **negative**)

Lack of **superoxide** leads to less ROS, RNOS, and HOCL.
Less destruction of bacteria and fungi.

Severe recurrent infections:
- **fungal**
- **bacterial pneumonia**

Myeloperoxidase (MPO) deficiency

Myeloperoxidase deficiency
(respiratory burst test **positive**)

Lack of **HOCL** leads to less destruction of **fungi.**
Normal destruction of bacteria.

Recurrent **fungal** infections with candida albicans.

Summary: fate of superoxide

Molecular oxygen + NADPH

NADPH oxidase

Superoxide

Superoxide dismutase

Hydrogen peroxide

plus chloride ions

Myeloperoxidase

Hypochlorous acid

Superoxide

Superoxide dismutase

Hydrogen peroxide

ferrous iron or superoxide

Non-enzymatic

Hydroxyl radical

Molecular oxygen + NADPH + Arginine

Inducible NO synthase

Superoxide

Nitric oxide

Non-enzymatic

RNOS

Thank you.

If you have questions, please email

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